

My Understanding of Machine Learning Types

Supervised Learning: I understand supervised learning as learning from examples that already have the correct answers. The model uses those examples to make predictions on new data.

Unsupervised Learning: This is when a model looks for patterns or groups in data without being told what the answers should be.

Reinforcement Learning: This is learning by trying different actions and getting feedback. The agent gradually learns which actions lead to better rewards.

My Analysis of the Wine Classification Project

Best performing model: Logistic Regression performed best, with 88.9% accuracy. The Decision Tree reached 83.3%.

Why do you think this model performed better?

In my opinion Logistic Regression worked better with the features we selected. The Decision Tree was limited to a depth of three, so it might not have picked up enough detail.

What would you try next to improve performance? I would try scaling the data and adding more features. My three-feature model scored 83.3%, so changing the features did not improve the result in this experiment.

Real-World Application Ideas

Industry of Interest: Cybersecurity.

ML Problem: Identifying phishing emails.

Type of ML: Supervised learning.

Data Needed: Examples of phishing and legitimate emails, including their messages, sender details, and links.

Key Learnings

Most important concept learned: I learned why we separate training and testing data. A model needs to make good predictions on new examples, not just the ones it learned from.

Most challenging part: The hardest part for me was understanding what precision and recall tell us beyond accuracy.

Questions for further exploration: I want to know how to choose better features and tell when a model is overfitting.

